

The influence of *Aspalathus linearis* (Rooibos) and dihydrochalcones on adrenal steroidogenesis: quantification of steroid intermediates and end pro...



The steroid hormone output of the adrenal gland is crucial in the maintenance of hormonal homeostasis, with hormonal imbalances being associated with numerous clinical conditions which include, amongst others, hypertension, metabolic syndrome, cardiovascular disease, insulin resistance and type 2 diabetes. *Aspalathus linearis* (Rooibos), which has been reported to aid stress-related symptoms linked to metabolic diseases, contains a wide spectrum of bioactive phenolic compounds of which aspalathin is unique. In this study the inhibitory effects of Rooibos and the dihydrochalcones, aspalathin and nothofagin, were investigated on adrenal steroidogenesis. The activities of both cytochrome P450 17 α -hydroxylase/17,20 lyase and cytochrome P450 21-hydroxylase were significantly inhibited in COS-1 cells. In order to study the effect of these compounds in H295R cells, a human adrenal carcinoma cell line, a novel UPLC-MS/MS method was developed for the detection and quantification of twenty-one steroid metabolites using a single chromatographic separation. Under both basal and forskolin-stimulated conditions, the total amount of steroids produced in H295R cells significantly decreased in the presence of Rooibos, aspalathin and nothofagin. Under stimulated conditions, Rooibos decreased the total steroid output 4-fold and resulted in a significant reduction of aldosterone and cortisol precursors. Dehydroepiandrosterone-sulfate levels were unchanged, while the levels of androstenedione (A4) and 11 β -hydroxyandrostenedione (11 β OH-A4) were inhibited 5.5 and 2.3-fold, respectively. Quantification of 11 β OH-A4 showed this metabolite to be a major product of steroidogenesis in H295R cells and we confirm, for the first time, that this steroid metabolite is the product of the hydroxylation of A4 by human cytochrome P450 11 β -hydroxylase. Taken together our results demonstrate that Rooibos, aspalathin and nothofagin influence steroid hormone biosynthesis and the flux through the mineralocorticoid, glucocorticoid and androgen pathways, thus possibly contributing to the alleviation of negative effects arising from elevated glucocorticoid levels. Copyright © 2011 Elsevier Ltd. All rights reserved.